The compose lemma, reduced

Pointwise Fejér/van-der-Corput→H5 composition with explicit constants

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Abstract

Round 378 attacks the named remainder COMPOSE of the terminal lane: the slope chain $M_3 \to N_3 \to N_2 \to$ Fejér energy $\to$ van der Corput $\to$ H5 $\to q_N < 1$ is, at the Fejér/vdC→H5→ q_N joints, a measured convention (r297 $\sigma \le -0.516$; r300 N_2-need 0.908; r324 atom-need 0.888), not a pointwise theorem. What is proved: every exact link as a finite inequality with explicit constants; the van der Corput bound at $H = \max(2, \lceil\sqrt{m}\rceil)$ as a constant-free theorem; the participation and Rényi identities; the kernel envelope; the two-branch reduction; and the log-complete m_0-solve (the r361 body dropped $2\log\log m$). What is *not* proved: a uniform bound on the ℓ^1 mass L_1 of the block field, a uniform Fejér/diagonal ratio R_0, a uniform edge scalar $|Z_{\mathrm{loc}}|$, and a family-uniform T1 constant C_K. The hydra-legal reduction is the pointwise implication

$$M_3 \le \phi(m) \ \wedge\ L_1 \le \Lambda \ \wedge\ S_F \le R_0\, D \ \wedge\ |Z_{\mathrm{loc}}| \le Z_0 < M \implies q_N < 1$$

for $m \ge m_0$, with ϕ, Λ, R_0, Z_0, M and m_0 named below. Hypothesis H5 (the pair-margin law) is a *sister* sufficient condition, not an extra hypothesis of the vdC route. No RH claim.

Status. Lemmas 1–4 are proved (finite algebra). Lemma 2 is proved (the H-rule). Lemma 3 is the Lean theorem `canonical_split`. Lemma 5 is proved (kernel majorant). Lemma 6 is proved (conditional on the floor and a C_K-cap). Proposition 1 names the reduced compose implication. Proposition 2 records the log-complete m_0 (Sol correction). The α^*-convention 0.888 vs. 0.908 is a slope dictionary, not a pointwise need. No RH claim.

1 The object

Fix a canonical window of constructive level-2 block count $m \ge 2$. Write $M = \sqrt{5/7}$ for the r243 floor, Z for the terminal drive readout, Z_{loc} for the edge-local piece of the r269 $F = 0.20$ split, and $P = (P_j)_{j<m}$ for a real block sequence (either the r287 level-2 border blocks, or the r298 difference field P_Δ). The Fejér energy at window H is

$$S_F = \frac{1}{H}\sum_t W_t^2 = \sum_{|h|<H}\Big(1 - \frac{|h|}{H}\Big)A(h),$$

with W_t the sliding H-window sums and A the autocorrelation of P. The frozen rule is $H = \max\big(2, \lceil\sqrt{m}\rceil\big)$. Write pref $= (m + H - 1)/H$, $D = \sum_j P_j^2$, $L_1 = \sum_j |P_j|$, $N_2 = L_1^2/D$ (participation), and $M_3 = \sum_j q_j^3$ with $q_j = |P_j|/L_1$ (the r324 convention; equivalently S_3 of r306). The Rényi numbers are $N_3 = M_3^{-1/2}$ and $N_2 = 1/\sum q_j^2$. The terminal cross-ratio is $q_N = \rho_{N-1}/(5/7)$. The r263 dictionary, measured on the 42-rung census and typed in

Lean as `pair_terminal_dictionary` (a `sorry`, transcription-blocked), reads $Z^2 = (5/7)\,q_N$. Hypothesis H5 is the named Prop `PairMarginLaw`: $|Z_{\rm loc}| + \text{pairBound} < M$.

Claim boundary: finite identities and a named reduction. Not evidence for or against the Riemann hypothesis.

2 Each link, typed

Exact finite theorems

Lemma 1 (Fejér window-sum and van der Corput)**.** *For every real sequence P of length m and every $H \in \{1, \dots, m\}$,*

$$\Big|\sum_j P_j\Big|^2 \le \frac{m+H-1}{H}\, S_F,$$

with $S_F = (1/H)\sum_t W_t^2 \ge 0$. Equality at $H = 1$. The prefactor m/H is false (witness $P = (1,1,1,1)$, $H = 2$: $14 < 16$).

This is v964-S0 (covering identity $\sum_t W_t = H \sum_j P_j$ plus Cauchy–Schwarz over $m + H - 1$ positions), re-proved in Fractions in the companion script.

Lemma 2 (The H-rule prefactor)**.** *At $H = \max(2, \lceil\sqrt{m}\rceil)$ one has* pref $\le \sqrt{m} + 1$ *for every $m \ge 1$.*

Lemma 3 (Canonical split)**.** $|Z| \le |Z_{\rm loc}| + |\sum P|$ *whenever $\sum P$ is the bulk remainder of the canonical extraction. This is `RH.PairBound.canonical_split` (no `sorry`).*

Lemma 4 (Participation and Rényi)**.** *$D \cdot N_2 = L_1^2$ identically. For a probability vector q, the Lagrange identities $2(S_2^2 - S_3) = \sum_{i,j} q_i q_j (q_i - q_j)^2 \ge 0$ and $2(S_3^2 - S_2 S_4) = \sum_{i,j} q_i^2 q_j^2 (q_i - q_j)^2 \ge 0$ give $N_2 \ge N_3 \ge N_4$, with equality exactly on the uniform vector. The r324 interpolation $M_3 \le q_{\max} M_2$ and $M_2 \le q_{\max}$ is the same algebra.*

Lemma 5 (Kernel envelope)**.** *The Fejér kernel*

$$F_H(\theta) = \Big|\sum_{j<H} e^{-ij\theta}\Big|^2 / H$$

satisfies $F_H(\theta) \le \min\big(H, 1/(H \sin^2(\theta/2))\big)$. Consequently $S_F \le R_{\rm env} D$ with $R_{\rm env}$ the F_H-weighted mean of $|\widehat{P}|^2$. The trivial bound $R_{\rm env} \le H$ is a theorem and is too coarse for composition (it demands $N_2 \gg m$).

Lemma 6 (T1-floor algebra)**.** *If $g_i \ge 3/8$ and $F_i = (mq_i/\log m)\, g_i \le C_K$ for every block, then $q_i \le (8/3) C_K (\log m)/m$ and*

$$M_3 \le \Big(\frac{8}{3} C_K \frac{\log m}{m}\Big)^2 = \frac{64}{9}\, C_K^2\, \frac{(\log m)^2}{m^2}.$$

The constant 8/3 is the r361 MED-CAP cap (saturated at χ_4 kz53). The lemma is a theorem of the floor *and* a C_K-cap; both remain named remainders of other rounds (V_3' and T1).

Measured, not theorems

- r298 identity $S_F = B(\omega, \omega) + B(\Delta, \omega + \beta)$ is exact (SATZ). The letter `TRANSFER_DOMINANT` ($S_F \approx B(P_\Delta, P_\Delta)$) is measured: the window main term is empty on the 42-rung ladder, not a cofinal law.

- r297 target $\sigma := \mathrm{slope}(S_F/M^2) \le \sigma^* = -0.516$ is a halves-log-slope on $N \in [142, 878]$. Truth $\sigma = -0.714$, margin 0.198. The identity $\sigma^* = 2(\mathrm{slope}(\mathrm{eps}_{c2}/M) - 0.21) - \mathrm{slope}(\mathrm{pref})$ is exact arithmetic on those slopes; it is *not* a pointwise bound on S_F.
- r300 need $\mathrm{slope}(N_2) \ge 0.908$ is the exact reparametrization $\mathrm{slope}(D) = 2\,\mathrm{slope}(L_1) - \mathrm{slope}(N_2)$ of $\mathrm{slope}(D) \le \sigma^*$, using the measured $\mathrm{slope}(L_1) = +0.196$. It is not a pointwise lower bound on N_2.
- r324 need $N_2 \gtrsim m^{0.888}$ is an atom-level slope target (r302 `ATOM_TARGET`), composed with a measured M_3 envelope and extrapolated to $m_0^* = 10^{59.6}$ (chain-honest bar 10^{238}). The power 0.888 is not the block-level Fejér need 0.908 (in N-units; $908/1002 \approx 0.906$ in m-units, r328B).
- r287 $\delta'(F2) = +0.31 > 0.21$ at $H = \lceil\sqrt{m}\rceil$ is a slope of the vdC *output*. The vdC inequality itself is Lemma 1 (classical, exact). The input size S_F is what was measured.
- The ratio B/D is measured median 1.29, falling; the structural majorant R_{env} is measured median 1.61, falling (`RATIO_BOUNDED_STRUCTURAL`). No absolute R_0 is proved.
- r306 $M_3 \le 1.069\,(\log m)^2/m^2$ holds on 57 large-gap rungs and fails on small-gap EXT3 (kz51 by 7.1×). It is a census, not a law.
- The 42-rung census $q_N < 1$ is a certificate: cheap branch 35/42 by the triangle, six exceptions by sealed phase bounds, kz15 by the r270 interval certificate (dps 640, reserve 0.0268). Dictionary image Z^2/M^2 has min margin 0.0195; the sealed ρ-margin is 0.0139. Not cofinal. On the same 42, vdC-F2 closes 38/42 (cheap 32/35) and the H5-sister pair bound closes 21/42.

Chain table

link	form	status	constant
Fejér id.	$S_F = (1/H)\sum W^2$	SATZ	—
vdC	$\lvert\sum P\rvert^2 \le \mathrm{pref}\, S_F$	SATZ (v964-S0)	$\mathrm{pref} = (m+H-1)/H$
H-rule	$\mathrm{pref} \le \sqrt{m}+1$	SATZ	$H = \max(2, \lceil\sqrt{m}\rceil)$
split	$\lvert Z\rvert \le \lvert Z_{\mathrm{loc}}\rvert + \lvert\sum P_{F2}\rvert$	SATZ (Lean)	—
part.	$D\,N_2 = L_1^2$	SATZ	—
Rényi	$N_2 \ge N_3 = M_3^{-1/2}$	SATZ	—
envelope	$F_H \le \min(H, 1/(Hs))$	SATZ; R_0 open	—
T1 alg.	$M_3 \le \big((8/3)C_K \log m/m\big)^2$	SATZ cond. $g \ge 3/8$, C_K	8/3
r298	$S_F = B(\omega,\omega) + B(\Delta, \omega+\beta)$	SATZ; TRANSFER measured	—
r297 σ^*	slope ≤ -0.516	convention	truth -0.714, margin 0.198
r300	$\mathrm{slope}(N_2) \ge 0.908$	convention	$2\cdot 0.196 + 0.516$
r324	$N_2 \ge m^{0.888}$	convention	ATOM_TARGET
r287 δ'	$+0.31$	measured output slope	input S_F measured
H5	$\lvert Z_{\mathrm{loc}}\rvert + \mathrm{pairBound} < M$	named Prop (sister)	H1–H4 proved
Dict	$Z^2 = (5/7)q_N$	named `sorry`	42/42 measured
Head	$q_N < 1$ on 42 rungs	certificate	q_Z-margin 0.0195
COMPOSE⁻	$M_3 \le \phi(m) \wedge$ (R)(L)(Z) $\Rightarrow \lvert Z\rvert < M$	SATZ reduced	ϕ as (1)

3 The pointwise composition

Combine Lemmas 1–3:

$$|Z| \le |Z_{\mathrm{loc}}| + \sqrt{\mathrm{pref} \cdot S_F}.$$

If the right-hand side is strictly less than M, then $|Z| < M$. Under the r263 dictionary this is $q_N < 1$. No slope enters.

Write $S_F \le R_0 D = R_0 L_1^2/N_2$ (Lemma 5 supplies the shape; R_0 is the remaining bound) and $N_2 \ge N_3 = M_3^{-1/2}$ (Lemma 4). Then

$$|Z_{\mathrm{loc}}| + \sqrt{\mathrm{pref}\, R_0}\, L_1\, M_3^{1/4} < M$$

is sufficient. Rearranging with $|Z_{\mathrm{loc}}| \le Z_0 < M$ and $L_1 \le \Lambda$ yields the M_3-cap

$$M_3 \le \phi(m) := \frac{(M - Z_0)^4}{\mathrm{pref}(m)^2\, R_0^2\, \Lambda^4}, \qquad \mathrm{pref}(m) \le \sqrt{m} + 1. \tag{1}$$

Asymptotically $\phi(m) \sim c_\phi\, m^{-1}$ with $c_\phi = (M - Z_0)^4/(R_0^2 \Lambda^4)$, i.e. the H5-sufficient exponent is $\beta = 1/2$ in the form $M_3 \le c_1 (\log m)^{-a}\, m^{-2\beta}$ ($a = 0$, $c_1 = c_\phi$, $2\beta = 1$), *provided* Λ and R_0 are $O(1)$.

If $L_1 \le C\, m^\gamma$, the same algebra gives $2\beta = 1 + 4\gamma$, so $\beta = 1/2 + 2\gamma$. The measured slope $\gamma \approx 0.196$ produces $\beta \approx 0.892$, which is the origin of the 0.888/0.908 convention: those numbers are slope-proxies for $1/2 + 2\gamma$, not independent targets. T1-floor (Lemma 6) supplies $\beta = 1$, which beats $1/2 + 2\gamma$ for every $\gamma \le 1/4$.

Proposition 1 (Compose, reduced)**.** *Let $M = \sqrt{5/7}$, $H = \max(2, \lceil \sqrt{m} \rceil)$, $\mathrm{pref} = (m + H - 1)/H$. On a canonical window of block count m, assume*

(R) *$S_F \le R_0\, D$ for a frozen $R_0 < \infty$,*

(L) *$L_1 \le \Lambda$ for a frozen $\Lambda < \infty$,*

(Z) *$|Z_{\mathrm{loc}}| \le Z_0 < M$,*

(M3) *$M_3 \le \phi(m)$ with ϕ as in* (1)*.*

Then $|Z| < M$. If the r263 dictionary holds, $q_N < 1$. If in place of (L) one has $L_1 \le C m^\gamma$ with $\gamma < 1/4$ and in place of (M3) the T1-floor bound of Lemma 6 with a frozen C_K, the same conclusion holds for all $m \ge m_0$ of Proposition 2.

Hypothesis H5 is not an extra premise: the pair-margin law is a sister sufficient condition ($|Z_{\mathrm{loc}}| + \mathrm{pairBound} < M$). The vdC route replaces `pairBound` by $\sqrt{\mathrm{pref}\, S_F}$. On the 42-rung head the two-branch certificate already closes $q_N < 1$ without compose (cheap triangle 35/42; six F2/phase certificates; kz15 exact-finite). Compose is the cofinal statement.

4 Log accounting and m_0

The r361 direct chain used CRIT $= 0.224 = 2(1 - 0.888)$ as the budget for $N_2 \ge m^{0.888}$: $(8/3)^2 C_K^2 (\log m)^2 \le m^{0.224}$, and solved $0.224 \log m \ge 2 \log((8/3) C_K)$, dropping $2 \log\log m$. The published thresholds $10^{10.0}$ ($C_K = 4.91$) and $10^{16.1}$ ($C_K = 23.70$) are therefore the log-free approximants. The complete inequality is

$$0.224\, \log m \ge 2 \log\left(\tfrac{8}{3} C_K\right) + 2 \log\log m. \tag{2}$$

Proposition 2 (Log-complete m_0)**.** *The unique positive solution of* (2) *(equivalently of $0.224\, t = 2 \log((8/3) C_K) + 2 \log t$, $t = \log m$) is*

$$\log_{10} m_0 \approx \begin{cases} 25.81 & \text{at the rule-conditional freeze } C_K = 4.91, \\ 32.85 & \text{at the measured census ceiling } C_K = 23.70 \end{cases}$$

(companion script: 25.81 / 32.85 by bisection; the Sol diagnosis of DCCXXXVIII). The r361 figures $10^{10.0}$ and $10^{16.1}$ are the log-free undershoots (complete-inequality ratios 0.4517 and

0.5351*, both* < 1*: those* m *do* not *meet* (2)*; Sol's* 0.0437/0.0271 *compared a different left-hand side). The asymptotic claim "every polylog* C_K *suffices against a fixed power* $m^{0.888}$*" remains correct; the finite head threshold was optimistic.*

Against the *pointwise* H5-need $\beta = 1/2$ (bounded L_1, R_0, Z_0), T1-floor with any frozen C_K is far stronger than necessary, and m_0 is modest: the inequality $(64/9)C_K^2(\log m)^2/m^2 \le c_\phi/m$ reduces to $m/(\log m)^2 \ge K$ with $K = (64/9)C_K^2/c_\phi$. The large published m_0 were an artefact of composing T1 against the slope-proxy $m^{0.888}$ rather than against (1).

5 The remaining lemma set

The compose implication of Proposition 1 is a theorem of finite algebra. The original COMPOSE task is therefore *reduced* to the following named remainders, each strictly smaller than a slope law.

id	status	statement
(R)	lemma	$S_F \le R_0 D$ with an absolute R_0 (the r300 envelope is a theorem; the spectral weight is not).
(L)	lemma	$L_1 \le \Lambda$, or $L_1 = O(m^\gamma)$ for some $\gamma < 1/4$.
(Z)	lemma	$\lvert Z_{\mathrm{loc}}\rvert \le Z_0 < M$ cofinally on the generic half (cheap branch: triangle; exceptions: finite certificates).
(T1)	lemma	a family-uniform $C_K < \infty$, or any $M_3 = O((\log m)^B/m^2)$.
(V_3')	lemma	the r374 profile remainder of MED-CAP (supplies 8/3).
(Dict)	named	$Z^2 = (5/7)\,q_N$ (Lean `sorry`; measured 42/42).
(Head)	certificate	$q_N < 1$ on the 42 rungs plus kz15 (v960, already sealed).

H5/`PairMarginLaw` is *not* on this list for the vdC route. It remains the named pair-form certificate route (H1–H4 Lean-proved; H5 open as a Prop). The two routes share the split and the dictionary.

The α^*-ambiguity is dissolved: 0.888 is `ATOM_TARGET` (atom participation, m-units); 0.908 is `NEFF_TARGET` (block participation, N-units); the pointwise H5-need with bounded L_1 is $\beta = 1/2$. None of the three is a theorem. Compose replaces all three by (1).

6 Machine check

The companion script `verify_compose_lemma.py` gates every numbered lemma on Fractions toys (G1–G11), the four $w = 9$ worlds (G20), χ_4 kz53 floor saturation (G21), the sealed 42-rung FRAME-A arithmetic (G22: two-branch set $\{15, 20, 22, 36, 38, 39, 52\}$, F2-split and reconstruction 42/42, cheap triangle 35/42, vdC-F2 38/42, H5-sister 21/42, dictionary-image min q-margin 0.0195), and the T1-floor implication on the sealed 181-row surface (G23: 89 A +8 B +42 χ_3 +42 χ_4; $N_2 \ge N_3$ and row-cap 181/181). 15/15 gates. Final line `COMPOSE LEMMA VERIFIED`. No RH claim.

References

[1] TFPT verification module `v964_lstar_coherence_census.py`, van der Corput S0.

[2] TFPT verification module `v966_l2_reduction_chain.py`, rounds r296–r300.

[3] TFPT verification module `v960_terminal_surface_closure.py`, 42-rung certificate.

[4] `rh/lean/RH/PairBound.lean`, H1–H4 and `PairMarginLaw`.